

Thanushkumar Sasinthra Thilipkumar

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Professional Experience:

Junior Software Engineer: Yapily, Oxford Circus

April 2026 – Present

- Led and executed the solo migration of database access controls across dev, staging, and production environments, implementing GCP IAM authentication via Terraform to replace manual credential logins for all engineering teams.
- Investigated and resolved a floating-point precision defect causing payment value discrepancies between raw Open Banking API responses and BigQuery-stored billing data, tracing and applying the fix across 2 microservices.

Associate Software Engineer, Data: EquiLend, Canary Wharf

Aug 2025 – April 2026

- Led the migration of streaming workloads from Confluent ksqldb to Apache Flink, driving requirements gathering, implementation planning, and successful deployment for an enterprise cluster rollout.
- Optimized critical security price file workflows in Apache Airflow, driving a 75% performance gain by cutting execution time from 2 hours to 30 minutes.
- Architected and deployed 10+ Airflow ingestion pipelines on Databricks, automating raw data processing and enabling reliable, production-ready analytics datasets.
- Engineered automated ingestion pipelines to extract and process daily reference rates from the New York FED website, ensuring 100% data availability for business-critical downstream financial applications
- Built an ETL pipeline using Airflow and Databricks to integrate Salesforce and database data, automating revenue calculations and improving accuracy.

Graduate Technology Analyst: EquiLend, Canary Wharf

Sep 2024 – Jul 2025

- Developed Java REST endpoints on the Quarkus framework using Mutiny for reactive programming.
- Refactored the REST API codebase to improve code quality and optimize PostgreSQL JSON queries with JSONB functions, resulting in cleaner architecture and better query performance
- Implemented custom metrics using Micrometer to enable observability across 6 microservices, integrating with Prometheus and Grafana for enhanced monitoring and insights.
- Refactored a monolithic stored procedure into two separate procedures to resolve stale security prices, ensuring up-to-date data display and improved performance.

Software Engineering Intern: NCR Corporation, Paddington

May 2023 – Aug 2023

- Built a Java data-import tool to streamline performance testing file generation; conducted Linux-based deployment testing across the software installation lifecycle.

Projects:

Agentic AI Portfolio Advisor: FastAPI | LangGraph | MongoDB Atlas | React.js | Cohere LLM

- Built a multi-agent financial advisor using LangGraph and MongoDB Atlas Vector Search with Voyage AI embeddings, deployed to AWS in one day at the MongoDB Hackathon.
- Exposed a FastAPI backend consumed by a React/Vite dashboard, using Cohere Command-R+ for response synthesis.

Movie Recommendation Web App: React.js | Flask | Python (Final Year Project)

- Designed and implemented a recommendation system using NLP techniques, exposing results through a Flask API and React frontend.

Weather App for Disabled people: React.js | JavaScript | HTML | CSS (Group Project)

- Developed an accessible React.js weather app with a 4-person team, integrating APIs tailored for disabled users.

Skills:

Backend & APIs: Spring Boot, Quarkus, REST API, Micrometer, Prometheus, Grafana

Languages: Java | Python | JavaScript | SQL

AI & LLMs: LangGraph, RAG, Cohere, Claude Code, Prompt Engineering

Data & Pipelines: Apache Airflow, Apache Flink, Databricks, Delta Lake, Kafka

Cloud & Infra: AWS (**Certified Solutions Architect**), GCP (IAM, Pub/Sub, CloudSQL), Terraform, Docker

Databases: PostgreSQL, MongoDB, Oracle

Education:

BSc (Hons) Computer Science (First Class), Queen Mary University of London

Sep 2021 – July 2024

Deep Learning & Neural Networks | Big Data Processing | Software Engineering | Computability & Algorithms

A-levels, City & Islington Sixth Form College

Nov 2019 – May 2021

Mathematics (A) | Electronics (A*) | Computer Science (B)